

# Modelling the behaviour of shape memory materials using machine learning approaches

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# Overview

- 1 Motivation/Aim
- 2 Shape memory materials and its applications
- 3 Shape memory effect and approaches for its behaviour modeling
- 4 A review of numerical implementation of SMA thermo-mechanical model using Return-Mapping algorithms
- 5 Case Study: Artificial Neural Network Model
- 6 Future work

# Motivation

The Shape Memory Materials (SMM) are stimuli-responsive smart materials with the ability to undergo a large recoverable deformation upon the application of an external stimulus like heat, electricity, light, magnetism, moisture, change in pH value.

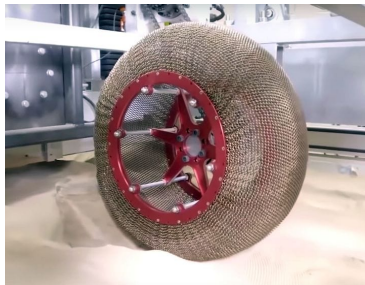


Figure: NASA's shape memory alloy wheel

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<sup>1</sup>Image credits: <https://patents.google.com/patent/US10449804B1/en>

Shape memory alloys (SMA) are capable of delivering advantageous solutions to a wide range of engineering-based problems. Implementation of these solutions, however, is often complicated because of:

- Hysteresis, non-linear thermo-mechanical behaviour of the materials
- Existing constitutive models inherently depend on the accurate characterization of the material, identifying values of material parameters
- Current characterization procedures necessitate extensive data collection and data processing, creating a high barrier of entry for shape memory alloy application.

# Aim

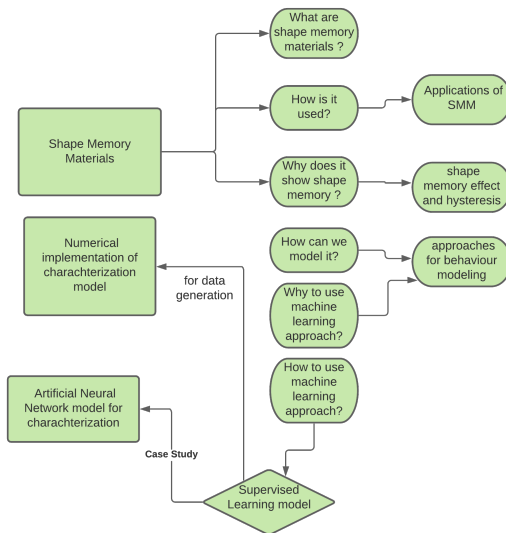
In this project,

I developed an understanding about the approaches for shape memory material behaviour characterization using machine learning

Conducted review of numerical implementation method for training data generation with the use of existing shape memory alloy constitutive model

Also I conducted a case study of supervised learning model developed to identify transformation temperatures and stress influence coefficients of shape memory alloy

# Project Layout



# How are SMMs used? | Applications

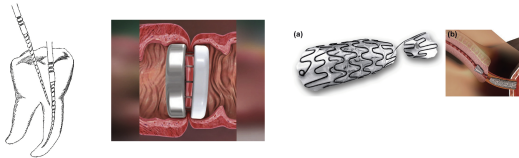


Figure: Various Bio-Medical Applications of SMA[3]

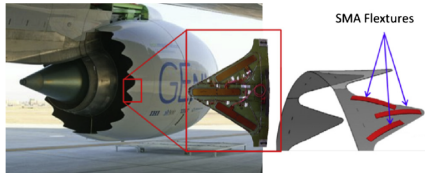


Figure: Boeing's variable geometry chevron[2]

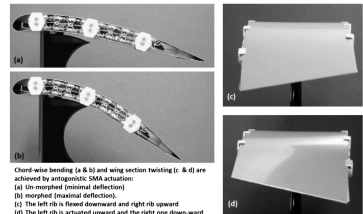


Figure: Wing Morphing with SMA Actuators[2]

# How are SMMs used? | SMA Actuator

Due to similarity in performing the operation of sensing, controlling and actuating, for the design of SMA based actuators MEMS based actuators are studied.

The key feature of a shape memory actuator is it uses only the phase transformation of the alloy for producing force. The strain path, or type of stroke depends on the processed shape of the actuating element

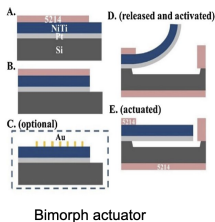


Figure: Micro Shape Memory Alloy Actuator[5]



# Shape memory effect and hysteresis

The phase transformation as seen in the figure 3.1 from martensite to austenite (heating) causes a decrease in length while the reverse process extends it back to its original length in case of using SMA wire for actuation.

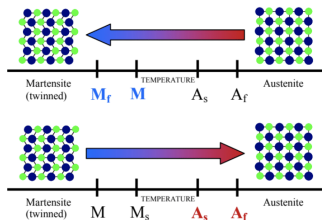


Fig. 1a Thermally Induced Phase Transformations

**Figure:** Thermally induced Phase transformation[4]

# Approaches for SMM behaviour modeling

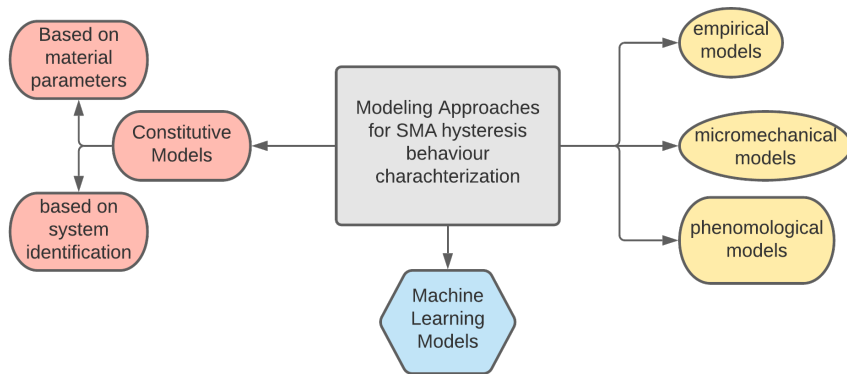
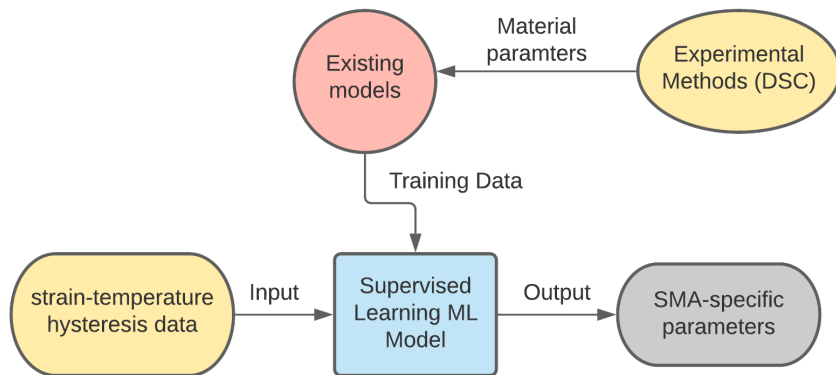


Figure: Modelling Approaches for SMM behaviour characterization

# Machine Learning based approach for SMA behaviour characterization



**Figure:** Supervised machine Learning model for shape memory material based actuators' behaviour characterization process summary diagram:

# Numerical Implementation of Thermo-mechanical Constitutive models using return mapping algorithms

To address this issue of Numerical Implementation of Thermo-mechanical constitutive models, authors Siddiq et al. proposed usage of return mapping algorithms

- Unified Thermo-mechanical constitutive model is implemented for the work by the authors with infinite strain constraint assumption
- Because these return mapping algorithms are used to integrate incremental constitutive equations for rate-independent inelastic materials; In case of SMA rate-independence and loading history dependence is observed thus this algorithms are useful and suited to be applied

# Closest point projection return mapping algorithm

The key feature of the closest point projection algorithm is it integrates the transformation equation for the correction using the backward euler integration rule ( $\lambda = 1$ ) resulting in non-linear algebraic equation which are solved using newton's method.

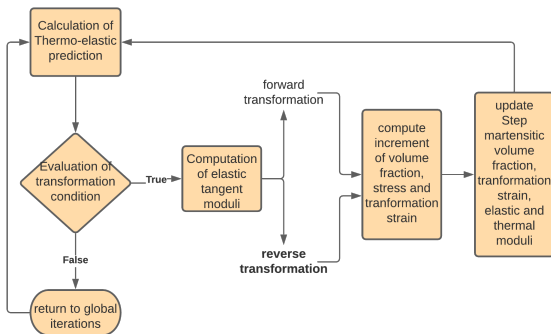
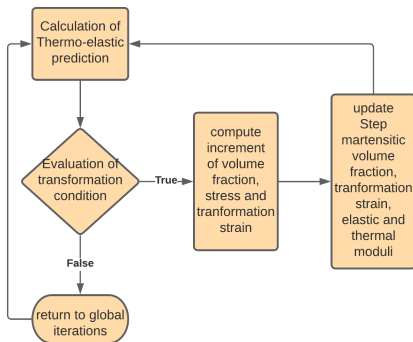


Figure:

# Convex cutting plane return mapping algorithm

- Integration of transformation correction is done in explicit manner with linearizing the consistency condition.
- The algorithm converges to final value in a quadratic rate but convergence is not guaranteed.
- No transformation residual is required to be computed.



# Comparison of both algorithms

## Closest point projection algorithm

- The quadratic convergence rate is guaranteed, it is unconditionally stable and first order accurate
- includes derivation of consistent closed form tangent moduli tensor preserving global quadratic convergence rate
- Involves high computation effort and complexity

## Convex Cutting Plane algorithm

- The forward transformation step is unconditionally stable while for the reverse transformation newton method is locally convergent
- Derivation of consistently closed form tangent moduli is not possible
- Fundamental advantages are simplicity and lower computational effort

# Case Study: Artificial Neural Network Model

ANN-based Approach developed by henrickson is capable of identifying material parameters the transformation temperature and stress influence coefficients

- Implementation of constitutive SMA model allows for the generation of strain temperature hysteresis plots using SMA parameters as inputs, the desired value of the trained ANN is to reverse this process.
- Accuracy of 87% is observed for transformation temperature and 73% with stress influence coefficients, with the use of 428 specimens for training.
- One of the key findings includes the implementation of the Taguchi method which allowed the development of a high-performance ANN model with better accuracy using 428 specimens compared to the same model trained with 4296 specimens without Taguchi method.



# Future work

- ① Developing supervised machine learning model based on the understanding gained through the project.
- ② Performing the data generation and implementation of the ML model
- ③ Tuning the ML model parameters, exploring ways to improve the accuracy
- ④ Exploring hybrid machine learning approach for the purpose of characterization

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# Thank you!